



Creating a Private Subnet

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The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation at the top reads 'VPC > Subnets > Create subnet'. The page title is 'Subnet settings' with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.' Below this, it indicates 'Subnet 1 of 1'. The 'Subnet name' section has a text input field containing 'Nextwork-Private-Subnet-5' and a note that the name can be up to 256 characters long. The 'Availability Zone' section has a dropdown menu set to 'No preference'. The 'IPv4 VPC CIDR block' section has a dropdown menu set to '10.0.0.0/24'. The 'IPv4 subnet CIDR block' section has a text input field set to '10.0.0.0/24' with a '256 IPs' indicator and a slider below it. At the bottom, there is a 'Tags - optional' section with a table. The table has two columns: 'Key' and 'Value - optional'. One tag is present with the key 'Name' and the value 'Nextwork-Private-Subnet-5'. There are 'X' buttons to delete the tag and a 'Remove' button.

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Nextwork-Private-Subnet-5
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/24

IPv4 subnet CIDR block
10.0.0.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional
Name	Nextwork-Private-Subnet-5

Remove



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is an isolated network within AWS where I can launch and organise my cloud resources. It is useful because it gives me full control over my network environment, allowing me to separate public-facing resources from private ones, control traffic through route tables, and apply multiple layers of security through security groups and network ACLs.

How I used Amazon VPC in this project

In today's project, I built a private subnet architecture inside my VPC. I created a dedicated route table with only a local route and no internet gateway connection, keeping my private subnet isolated from the internet. I then set up a custom network ACL starting with deny-all rules, giving me explicit control over what traffic reaches my private subnet. Together, these components ensure sensitive resources like databases are protected while still being able to communicate with other resources inside my VPC.

One thing I didn't expect in this project was...

I didn't expect the default NACL to allow all traffic by default. I assumed that a newly created subnet would be locked down automatically, but AWS leaves the default NACL wide open to help beginners get started. Creating a custom NACL was what actually gave me the deny-all starting point I needed for a private subnet. That difference between default and custom NACLs was a key learning moment.



This project took me...

This project took me 90 minutes.



Private vs Public Subnets

Private subnets have no route to an internet gateway, keeping them isolated and secure. This makes them ideal for resources such as databases and backend servers that don't need to be publicly accessible. Public subnets, on the other hand, have a route to an internet gateway, making them accessible from the internet and suitable for resources like website frontends that users need to access directly.

Private subnets exist because not every resource in a VPC should be publicly accessible. A good example of this would be my backend database for my website hosted on EC2. Users need access to the website's frontend, but never to the database behind it. Keeping the database in a private subnet ensures it has no route to the internet gateway, protecting sensitive data from public exposure. Private subnet resources are still able to communicate with other resources inside the VPC, so the EC2 instance and database can still interact securely.

My private and public subnets cannot share the same CIDR block because it would mean their IP address ranges are overlapping. It's like two houses in a suburb with the same addresses and postal code. A visitor would have a difficult time navigating to the right one. Every subnet in a VPC must have a unique CIDR block so that traffic is routed correctly and there are no conflicts, regardless of whether the subnets are public or private.



VPC > Subnets > Create subnet

Subnet Settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Nextwork-Private-Subnet-5

The name can be up to 256 characters long.

Availability Zone [info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block [info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/24

IPv4 subnet CIDR block

10.0.0.0/24 256 IPs

< > ^ v

Tags - optional

Key

Value - optional

Q Name

Q Nextwork-Private-Subnet-5

Remove



A dedicated route table

By default, my private subnet is associated with the main route table (the default route table that AWS automatically creates with every VPC). Any subnet not explicitly assigned to a custom route table falls back to the main route table. In this step, I manually associated my private subnet with my private route table instead, ensuring it has only a local route and no path to the internet.

I'm setting up a new route table specifically for my private subnet because I need dedicated control over its traffic rules. The main route table is shared by default, so any changes to it could accidentally affect other subnets. By creating a separate private route table with only a local route and no connection to an internet gateway, I can ensure that my private subnet stays isolated from the internet regardless of what happens to other routes in my VPC.

My private subnet's route table only has one route, which is a local route that directs all traffic within the VPC's CIDR range to stay internal. There is no route to an internet gateway, meaning that traffic can only flow between resources inside my VPC. This is what keeps the private subnet isolated from the public internet.



Route table rtb-0bedc2eab7ea2c0e8 | Nextwork-Private-Route-Table-5 was created successfully.

rtb-0bedc2eab7ea2c0e8 / Nextwork-Private-Route-Table-5

Actions

Details

Route table ID
rtb-0bedc2eab7ea2c0e8

VPC
vpc-09950d8505193146f | Nextwork-VPC-5

Main
No

Owner ID
479887546653

Explicit subnet associations
-

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	Create Route Table



A new network ACL

By default, my private subnet is associated with the default network ACL, which allows all inbound and outbound traffic via Rule 100. This is too permissive for a private subnet holding sensitive resources and that is why I am creating a custom NACL with deny-all rules as a starting point, allowing me explicit control over what traffic is allowed in and out.

I set up a dedicated network ACL for my private subnet because the default NACL allows all inbound and outbound traffic via Rule 100. This is too permissive for a private subnet holding sensitive resources. A custom NACL with deny-all rules as a starting point, allows me explicit control over what traffic is allowed in and out.

My new custom NACL has only one rule on the both inbound and outbound side. Rule *, which denies all traffic from all IP addresses. There are no allow rules yet, meaning all traffic to and from my private subnet is completely blocked until I explicitly add rules above Rule *.



You have successfully updated subnet associations for acl-0404b611f88a9996f / Nextwork-Private-NACL-5.

Details

Network ACLs (1/3) Info

Actions Create network ACL

Find Network ACLs by attribute or tag

	Name	Network ACL ID	Associated with	Default	VPC ID	Inbo
☐	-	acl-06f6b0608c7e9ea57	6 Subnets	Yes	vpc-0ce89e2934bb6614c	2 Int
☐	-	acl-0c4584eb14f048db8	-	Yes	vpc-09950d8505193146f / Nextwork-V...	2 Int
☒	Nextwork-Private-N...	acl-0404b611f88a9996f	subnet-0973a748aca3c1def / Nextwork-Private...	No	vpc-09950d8505193146f / Nextwork-V...	1 Int

acl-0404b611f88a9996f / Nextwork-Private-NACL-5

Details Inbound rules Outbound rules Subnet associations Tags

Inbound rules (1)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny



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